(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JUNE 2024

Paper Code: ES-108 ***Subject: Electrical Science***

Time: 3 Hours ***Maximum Marks: 60***

Note: Attempt five questions in all including question no.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Attempt **all** **(4x5=20)**

(a) An alternating current is given by i = 14.14 sin 377 t. Find the time taken for the current to reach 10 Ampere for the first time after passing through zero value.

(b) What are the different losses in transformers?

(c) What is the meaning of time domain analysis of first order R-L and R-C circuits?

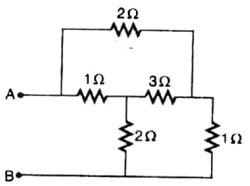
(d) What are the two difference between induction and synchronous motor?

(e) What are the different torques associated with measuring instruments?

UNIT-I

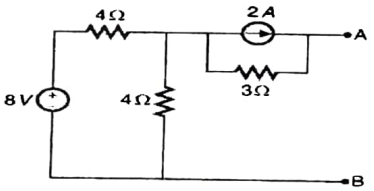
Q2 (a) Find the equivalent resistance R_{AB} between terminal A & B in Figure1 by Star-Delta transformation. **(6)**

Figure-1



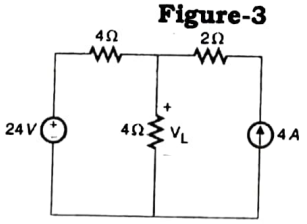
(b) Find Thevenin's equivalent of the circuit shown in Figure 2. **(4)**

Figure-2



Q3 (a) Find the voltage V_L in the circuit shown in Figure 3 using the principle of superposition. **(3)**

Figure-3



(b) State and prove maximum power transfer theorem. **(7)**

UNIT-II

Q4 (a) An inductor coil takes 10 A and dissipates1000 W when fed from 250 V 25Hz supply. Find the impedance, resistance, inductance, and power factor of the coil. **(6)**

(b) Explain the series resonance phenomenon in the series R-L-C circuit and derive the bandwidth formula of the circuit. **(4)**

P.T.O.

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MID TERM EXAMINATION

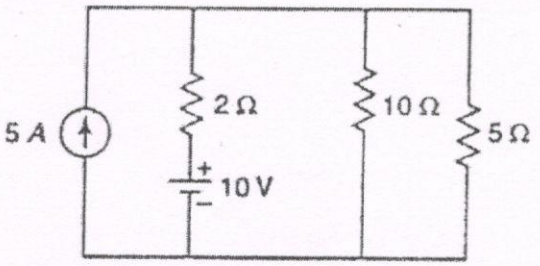
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

2nd Semester, April 2024

Paper Code: ES-108 **Subject: ELECTRICAL SCIENCE**

Time: 1½Hrs. **Max. Marks: 30**

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

Q.No.	Questions	Max. Marks	CO(s)
1(a)	What is the response of series RC circuit with AC supply. Explain with phasor diagram.	2	CO2
1(b)	If the waveform of a current has form factor 1.2 and peak factor 1.7, find the average and rms value of the current if the maximum value of the current is 100 A?	2	CO2
1(c)	Define true power and complex power?	2	CO2
1(d)	Write the expression for transforming a star connected network in a delta network.	2	CO1
1(e)	Differentiate bilateral and unilateral elements with examples.	2	CO1
2(a)	State and proof the maximum power transfer theorem having efficiency 50% with appropriate diagram and expression.	5.0	CO1
2(b)	Find the current through the 5Ω resistor using Thevenin's theorem. 	5.0	CO1
3(a)	A circuit having a resistance of 4Ω, an inductance of 0.5 H and a variable capacitance in series, is connected across a 100 V, 50 Hz supply. Calculate (i) the capacitance to give resonance (ii) the voltage across the inductance and the capacitance (iii) the Q factor of the circuit.	5.0	CO2
3(b)	Derive the expression for the resonant frequency and bandwidth of a series R-L-C circuit. Draw the phasor diagram for the same.	5.0	CO2
4(a)	Find i_1 and i_2 of the circuit shown in the following Fig. using superposition theorem.	5.0	CO1

PTO

1ST SEMESTER 2024

(Please write your Exam Roll No.) *Exam R. No. .....*

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] JANUARY 2024

Paper Code: ES-101 ***Subject: Programming in 'C'***

Time: 3 Hours ***Maximum Marks:*** 60

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Assume missing data if any.

Q1 Attempt all questions:- **(2x6=12)**

(a) What is recursion? Differentiate direct and indirect recursion?

(b) What is the return value of f(p,p), if the value of p is initialized to 5 before the call? Note that the first parameter is passed by reference, whereas the second parameter is passed by value.

```
int f (int &x, int c)
{
    c = c - 1;
    if (c==0) return 1;
    x = x + 1;
    return f(x,c) * x;
}
```

(c) Discuss the structure of the C program with examples.

(d) What is printed by the print statements in the program P1 assuming call by reference parameter passing?

```
Program P1()
{
  x = 10; y = 3;
  func1(y,x,x);
  print x; print y;
}
func1(x,y,z)
{
  y = y+4;
  z = x+y+z;
}
```

(e) Write a C program to print the Fibonacci series.

(f) Explain file handling. Differentiate sequential and random-access files.

UNIT-I

Q2 (a) What is an operator? Explain different types of operators. Discuss arithmetic, relational, logical, bitwise, assignment and comma operators in detail. **(6)**

(b) Explain storage classes. Elaborate auto, extern, static and register storage classes with the help of examples. **(6)**

Q3 (a) What is a token? Explain different types of tokens. Discuss keywords, identifiers, constants, and special characters in detail with examples. **(3)**

(b) What is a variable? Discuss the scope, visibility and lifetime of a variable. **(3)**

(c) What is the difference between call by value and reference? Write a C program to elaborate on their difference. **(6)**

P.T.O.

ES-101

UNIT-II

Q4 (a) What are jump statements? Discuss control structure. Explain different types of control structures. Elaborate the difference between while and do while loop control structure with the help of C program. **(6)**

(b) What is a 2D array? Explain the declaration and initialization of a two-dimensional array. Write a C program to initialize and print values using the 2D array. **(6)**

Q5 (a) What is a function? Explain elements of functions. Discuss different types of functions. Write a C program to elaborate on function definition, function call and function prototype. **(6)**

(b) What is a string? How string can be declared and initialized? Discuss the following string handling functions with examples. **(6)**

(i) strlen() (iv) strcpy()
(ii) strrev() (v) strspn()
(iii) strcat() (vi) strpbrk()

UNIT-III

Q6 (a) Explain file handling. What is the need for file handling? Discuss file access modes for text files: r, w, a, r+, w+, a+ and binary files: rb, wb, ab, r+b, w+b, a+b. **(6)**

(b) What is a pointer? How to declare and initialize a pointer? Discuss the significance. **(6)**
of * & operator. Write a C program to add two variables using Pointer.

Q7 (a) Explain structure and union? What is the difference between structure and union? Write a C program to elaborate on their difference. **(6)**

(b) Discuss file handling functions for input-output operations. Explain file positioning random access functions. Differentiate text and binary files. **(6)**

UNIT-IV

Q8 (a) What is a header file? Explain library functions of following header files in detail. **(6)**

(i) stdio.h (iv) ctype.h
(ii) stdlib.h (v) math.h
(iii) conio.h (vi) string.h

(b) What is binary search? Search item 23 from the following sorted data elements using binary search. **(6)**

2, 5, 8, 12, 16, 23, 38, 56, 72, 91

Q9 (a) What is insertion sort? Write a C program to implement insertion sort. **(6)**

(b) Explain selection sort. Sort the following numbers using selection sort. **(6)**

77, 33, 44, 11, 88, 22, 66, 55

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(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] JANUARY 2024

Paper Code: ES-107 **Subject: Electrical Science**

Time: 3 Hours **Maximum Marks: 60**

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Attempt all questions:-

(a) Distinguish between dependent and independent sources. How do you transform a voltage source into current source? **(2)**

(b) Show that power consumed in a purely inductive circuit is zero when sinusoidal voltage is applied across it. **(2)**

(c) Why a series resonant circuit is called as an acceptor circuit and a parallel resonant circuit as a rejector circuit? **(2)**

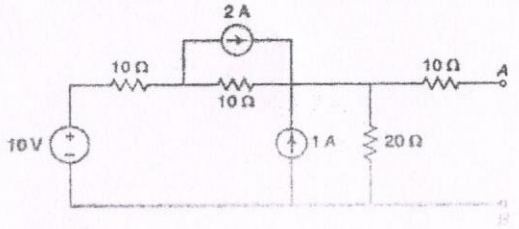
(d) Why is induced emf in a dc motor called the back or counter emf? **(2)**

(e) What is the difference between an ideal and practical transformer? **(2)**

(f) What type of damping is provided in PMMC instruments? **(2)**

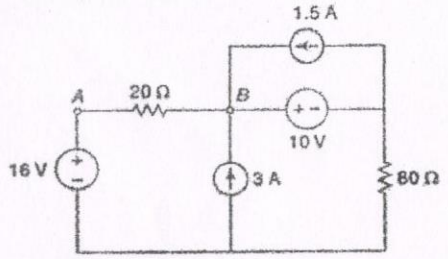
UNIT-I

Q2 (a) Find the Thevenin's equivalent circuit at terminals AB for the given network and also determine the power dissipated in a 5 Ω resistor connected between A and B. **(7)**



(b) Derive the expression for converting a star network to a delta equivalent network. (5)

Q3 (a) Using the superposition theorem, find the voltage across the 20 Ω resistor of the given circuit. **(7)**



(b) State and explain Norton's theorem. Show that Thevenin's and Norton's theorems are dual to each other. **(5)**

P.T.O.

UNIT-II

Q4 (a) A series circuit consists of a capacitor and a coil takes a maximum current of 0.314 A at 200 V, 50 Hz. If the voltage across the capacitor is 300 V at resonance. Determine the capacitance, inductance, resistance and Q of the coil. **(7)**

(b) Define parallel resonace with suitable phasor diagram. Calculate the resultant current and quality factor at resonance in terms of the pameters of a circuit. **(5)**

Q5 (a) A choke coil having a resistance of 10 Ω and inductance of 0.05 H is connected in series with a condenser of 100 μF. The whole circuit has been connected to 200 volt, 50 Hz supply. Calculate: **(7)**

(i) Impedance (ii) Current (iii) Power factor (iv) Power input (v) Apparent and Reactive power of circuit.

(b) Define quality factor in a series R-L-C circuit. Determine the half power frequencies in terms of quality factor and the resonant frequency for series R-L-C circuit. **(5)**

UNIT-III

Q6 (a) A 220 V, dc shunt motor takes a line current of 20 A. Resistance of shunt field winding is 200 Ω and resistance of the armature is 0.3 Ω. Find the armature current and back emf. **(7)**

(b) Derive the emf equation of a dc generator. Draw and expain different types of dc generator with suitable diagram. **(5)**

Q7 (a) A 3-phase, 50 Hz induction motor has 6 poles and operates with a slip of 5% at a certain load. Determine. **(7)**

(i) The speed of the rotor with respect to the stator.

(ii) The frequency of rotor current

(iii) The speed of the rotor magnetic field with respect to rotor.

(iv) The speed of the rotor magnetic field with respect to stator.

(b) What are the various methods of starting the single phase induction motor? Explain them in brief with relative advantages and disadvantages. (5)

UNIT-IV

Q8 (a) In a 25 KVA, 2000/200 V transformer the iron and full load copper losses are 350 W and 400 W respectively. Find the efficiency at unity power factor at (a) full load (b) half load. Determine the load for maximum efficiency. **(7)**

(b) Explain the working principle of Attraction type and Repulsion type moving iron instruments with the help of suitable diagrams. **(5)**

Q9 (a) A single phase transformer has a core whose cross -sectional area is 150 cm^2 ; operates at a maximum flux density of 1.1 Wb/m^2 from a 50 Hz supply. If the secondary winding has 66 turns, determine the output in KVA when connected to a load of 4 Ω impedance. Neglect any voltage drop in the transformer. **(7)**

(b) Draw the constructional features of transformer and explain the working in detail. Also derive the emf equation of transformer. **(5)**

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